

Three-Engine Architecture Reference

SCOUT SPEC v3.1 | SENTINEL v2.1 | APOLLO Analysis

Complete Signal Catalogs, Scoring Logic & Cross-Engine Comparison

All Pools, Timeframe Weights, Thresholds & Helper Functions

Table of Contents

Engine 1: SCOUT SPEC (v3.1)	1
1.1 Scoring Formula	1
1.2 Timeframe Architecture	1
1.3 Pool A: RSI Recovery (45 pts, 10 signals)	1
A1: RSI21 Reversal (max 7.5 pts, Gradual Power-Curve)	2
A2: RSI36 Reversal (max 3.0 pts, Gradual Power-Curve)	2
A3: RSI56 Reversal (max 2.0 pts, Gradual Power-Curve)	2
A4: Multi-Speed RSI Divergence (max 3.5 pts, Fire-and-Hold)	2
A5: RSI21 > RSI36 Crossover Spread (max 7.5 pts, Gradual)	2
A6: RSI21 > RSI56 Crossover Spread (max 5.0 pts, Gradual)	2
A7: RSI36 > RSI56 Crossover Spread (max 3.0 pts, Gradual)	3
A8: RSI Recovery Momentum (max 7.0 pts, Gradual)	3
A9: Cross-TF Recovery Alignment (max 4.0 pts, Binary Stepped)	3
A10: Multi-TF Trough Agreement (max 3.0 pts, Binary Stepped)	3
1.4 Pool B: Price Structure (30 pts, 5 signals) - v3 Gradual Redesign	3
B1: TP28 Convergence (max 3.5 pts, Gradual, 4H+Daily)	3
B2: WC50 Convergence (max 3.5 pts, Gradual, 4H+Daily)	3
B3: Price to TP28 Proximity (max 9.5 pts, Gradual, All TFs)	3
B4: TP28-WC50 Spread (max 6.0 pts, Gradual, All TFs)	4
B5: Price vs WAP21 (max 7.5 pts, Gradual, Daily-Only)	4
1.5 Pool C: Confirmation (12 pts, 3 signals)	4
1.6 Bonus Pools	4
1.7 Classification & Entry Thresholds	5
1.8 Trough Detection & Recovery Window	6

Engine 2: SENTINEL (v2.1)	7
2.1 Scoring Formula	7
2.2 Unique Feature: 4-Timeframe Architecture	7
2.3 Unique Feature: Scout Gating	8
2.4 Unique Feature: Multi-TF OR-Gate Cycle Activation	8
2.5 Pool A: RSI Recovery (55 pts, 8 signals x 3 TFs)	8
2.6 Pool B: Price Structure (35 pts, 4 signals x 3 TFs)	9
2.7 Pool C: Confirmation (37 pts, 7 signals)	9
2.8 Bonus Pools & Classification	10
Engine 3: APOLLO Analysis Engine	11
3.1 Scoring Formula	11
3.2 Unique Feature: Cross-Timeframe Trough Matching	11
3.3 Unique Feature: Recovery Momentum Windows	12
3.4 Pool A: Trough & Recovery Momentum (42 pts, 6 signals)	12
3.5 Pool B: Price Recovery & Structure (25 pts, 5 signals)	12
3.6 Pool C & Bonus Pools	13
3.7 Entry, Exit & Classification	14
Cross-Engine Comparison	15
Signal Type Taxonomy	15

Note: Right-click the TOC and select "Update Field" to refresh page numbers.

Engine 1: SCOUT SPEC (v3.1)

The SCOUT SPEC engine is the primary entry-scoring system designed for early detection of recovery opportunities following RSI troughs. It uses a multi-pool, multi-timeframe scoring architecture with 25 signals across 5 pools, producing a total score that drives trade entry decisions. Version 3.1 introduced gradual scoring across all Pool B signals to eliminate binary flip-flop behavior.

1.1 Scoring Formula

$$\text{Total Score} = \min(90, \text{Pool_A} + \text{Pool_B} + \text{Pool_C}) + \text{Bonus_A} + \text{Bonus_B}$$

Component	Max Points	Signals	Scoring Nature
Pool A (RSI Recovery)	45	10	Gradual power-curve + spread-based
Pool B (Price Structure)	30	5	Gradual / distance-based (v3 redesign)
Pool C (Confirmation)	12	3	Binary fire-and-hold + gated
Bonus A (Structural)	12	3	Binary stepped / composite
Bonus B (Uptrend Correction)	13	5	Gated by BB1 (uptrend confirmed)
Core Cap	90	20 signals	$\min(90, A+B+C)$
Grand Maximum	115	25 signals	$90 + 12 + 13$

Table 1: SCOUT SPEC scoring structure

1.2 Timeframe Architecture

The engine operates on three timeframes (4H, Daily, Weekly) with explicit per-signal weights. Weekly carries the highest weight for most Pool A signals, reflecting its importance in confirming trend reversals. Pool B uses 4H and Daily for convergence/spread signals and all three TFs for proximity. The WAP21 signal (B5) is Daily-only.

1.3 Pool A: RSI Recovery (45 pts, 10 signals)

Pool A forms the backbone of the SCOUT scoring system, measuring how effectively RSI indicators are recovering from trough conditions. Eight of the ten signals use gradual

scoring that ramps smoothly as recovery progresses, eliminating the binary on/off behavior that plagued earlier versions.

A1: RSI21 Reversal (max 7.5 pts, Gradual Power-Curve)

Measures recovery from the cycle-low RSI21 using a power-curve formula: $\text{recovery_ratio} = (\text{current_RSI21} - \text{cycle_low}) / (100 - \text{cycle_low})$, then $\text{points} = \text{max_pts} * \min(1.0, \text{recovery_ratio}^{0.7})$. The 0.7 exponent means early recovery earns points faster than linear, rewarding the initial bounce more heavily. Applied across all three timeframes with weights: 4H=1.5, D=2.5, W=3.5.

A2: RSI36 Reversal (max 3.0 pts, Gradual Power-Curve)

Identical power-curve logic applied to RSI36. TF weights: 4H=0.5, D=1.0, W=1.5. Captures medium-term RSI recovery.

A3: RSI56 Reversal (max 2.0 pts, Gradual Power-Curve)

Same power-curve on RSI56, the slowest RSI period. TF weights: 4H=0.5, D=0.5, W=1.0.

A4: Multi-Speed RSI Divergence (max 3.5 pts, Fire-and-Hold)

Uses a 30-bar lookback split into two 15-bar halves. Finds the local price minimum in each half. Requires the recent price low to be lower than the prior price low (price lower-low). Then checks RSI21, RSI36, and RSI56 at those two price lows for bullish divergence (RSI making a higher-low while price makes a lower-low). Requires at least 2 RSI periods showing divergence to score (2 periods = 60% weight, 3 periods = 100%). Fire-and-hold: once triggered, persists until next trough. TF weights: 4H=0.5, D=1.0, W=2.0.

A5: RSI21 > RSI36 Crossover Spread (max 7.5 pts, Gradual)

Measures HOW FAR RSI21 is above RSI36, not just whether it is above. Formula: $\text{scale} = \text{clamp}((\text{RSI21} - \text{RSI36}) / 4.0, 0, 1)$. A 4-point spread range was calibrated from the 75th percentile of observed spreads. Linear ramp from 0 spread to 4+ spread. TF weights: 4H=1.5, D=2.5, W=3.5.

A6: RSI21 > RSI56 Crossover Spread (max 5.0 pts, Gradual)

Same spread logic for the wider RSI21-RSI56 pair: $\text{scale} = \text{clamp}((\text{RSI21} - \text{RSI56}) / 7.0, 0, 1)$. The 7.0 range reflects the naturally wider spread between fast and slow RSI. TF weights: 4H=1.0, D=1.5, W=2.5.

A7: RSI36 > RSI56 Crossover Spread (max 3.0 pts, Gradual)

Spread logic for the medium-slow pair: $\text{scale} = \text{clamp}((\text{RSI36} - \text{RSI56}) / 3.0, 0, 1)$. Tightest range at 3.0 points since RSI36 and RSI56 track more closely. TF weights: 4H=0.5, D=1.0, W=1.5.

A8: RSI Recovery Momentum (max 7.0 pts, Gradual)

Measures the 3-bar rate of change in RSI21: $\text{momentum} = (\text{RSI21_current} - \text{RSI21_3bars_ago}) / 3$. Only scores if momentum exceeds 0.5 (positive recovery). Normalized: $\text{scale} = \text{clamp}((\text{momentum} - 0.5) / 2.0, 0, 1)$. TF weights: 4H=1.5, D=2.5, W=3.0.

A9: Cross-TF Recovery Alignment (max 4.0 pts, Binary Stepped)

Counts how many timeframes simultaneously show positive A8 momentum. 2 TFs = 2.0 pts; 3 TFs = 4.0 pts; 0-1 TFs = 0. Rewards multi-timeframe alignment of recovery momentum.

A10: Multi-TF Trough Agreement (max 3.0 pts, Binary Stepped)

Counts how many timeframes are currently in an active recovery window. 2 TFs = 1.5 pts; 3 TFs = 2.5 pts. Confirms that the trough was a multi-timeframe event, not a single-TF blip.

1.4 Pool B: Price Structure (30 pts, 5 signals) - v3 Gradual Redesign

Pool B was completely redesigned in v3 to eliminate binary scoring. All signals now use distance-based linear ramps, ensuring scores change gradually as price structure evolves. No signal flips from 0 to maximum on a single bar.

B1: TP28 Convergence (max 3.5 pts, Gradual, 4H+Daily)

Uses `_convergence_score()`: measures how tightly bunched the TP28 (Typical Price 28-bar moving average) values are over the last 10 bars. Coefficient of variation (CV) is calculated as $(\text{max} - \text{min}) / |\text{mean}| \times 100$. Linear ramp: CV = 0% = full points; CV >= 15% = 0 points. Requires at least 6 non-NaN values. TF weights: 4H=2.0, D=1.5.

B2: WC50 Convergence (max 3.5 pts, Gradual, 4H+Daily)

Same convergence logic applied to WC50 (Wilder's Close 50-bar moving average). Flat WC50 indicates reduced volatility and price consolidation, a prerequisite for directional moves. TF weights: 4H=2.0, D=1.5.

B3: Price to TP28 Proximity (max 9.5 pts, Gradual, All TFs)

Uses `_proximity_score()`: linear ramp measuring how close price is to TP28. If price is at or above TP28: full points immediately. If below: $\text{dist_pct} = (\text{TP28} - \text{Close}) / \text{TP28} \times 100$; $\text{scale} = 1.0 - (\text{dist_pct} / 12.0)$. At 12% below TP28, score reaches zero. This provides the strongest single signal in Pool B. TF weights: 4H=2.5, D=3.5, W=3.5.

B4: TP28-WC50 Spread (max 6.0 pts, Gradual, All TFs)

Uses `_spread_score()`: measures how much TP28 has pulled above WC50. $\text{spread_pct} = (\text{TP28} - \text{WC50}) / |\text{WC50}| \times 100$. Linear ramp: 0% spread = 0 points; 10%+ spread = full points. A positive and growing spread indicates that the shorter-term average is leading the longer-term average upward. TF weights: 4H=1.5, D=2.5, W=2.0.

B5: Price vs WAP21 (max 7.5 pts, Gradual, Daily-Only)

Uses `_price_wap_score()`: computes a 21-bar linearly weighted average of close prices (recent bars weighted more heavily). Two-component scoring: Distance component (60% weight) measures how far price is above WAP, with a 5% above-WAP ceiling for full distance points and a near-zone (-3% to 0%) providing partial credit. Persistence component (40% weight) counts consecutive bars where $\text{Close} > \text{WAP} \times 0.98$ (2% tolerance), scaling linearly over 21 bars. This signal directly addresses the need for price-based weightage in the scoring system.

1.5 Pool C: Confirmation (12 pts, 3 signals)

Pool C provides the final confirmation layer. These signals are mostly binary, acting as gates that confirm the recovery is genuine before a trade entry is considered.

Signal	Max Pts	Type	Logic
C1: Weekly Divergence	5.0	Binary fire-and-hold	Weekly-only, 20-bar lookback, price LL + RSI21 HL
C2: D-RSI21 > 50	3.0	Binary live/die	Daily RSI21 above 50
C3: Cross-TF Boost	4.0	Binary gated	Requires ≥ 2 TFs with crossovers active AND B3 score > 0

Table 2: SCOUT SPEC Pool C confirmation signals

1.6 Bonus Pools

Signal	Max Pts	Type	Logic
BA1: Volume Surge	5.0	Binary stepped	Vol/AvgVol: >1.5x=2, >2.5x=4, >3.5x=5
BA2: Support Bounce	3.0	Binary stepped	>=15% decline from 180-bar peak; proximity to 85% level
BA3: Trend Quality	4.0	Composite	MA structure + rising lows + correction compression + RS proxy
BB1: Uptrend Confirmed	2.0	Binary gate	MA200 rising AND Close > MA200 (prerequisite for BB2-BB5)
BB2: Correction Depth	5.0	Binary stepped	From 60-bar high: 25-35%=3, 35-45%=4, 45%+=5
BB3: RSI Turning Positive	2.5	Per-TF live/die	Any RSI crossing above 50; 4H=0.5, D=1.0, W=1.0
BB4: Price Above Support	2.0	Binary	Close > 85% of 180-bar structural peak
BB5: Volume on Recovery	1.5	Binary	>=3 up-days with avg up-day vol >= 1.2x overall avg

Table 3: SCOUT SPEC bonus pool signals

1.7 Classification & Entry Thresholds

Score Range	Classification	Action
0 - 49	No Entry	No action
50 - 64	Pilot Buy	Small position, 25-30% sizing
65 - 79	Standard Entry	Standard position
80 - 89	Full Entry	Full position
90+	Full Strong	Maximum position (rare)

Table 4: SCOUT SPEC classification tiers

Thresholds are calibrated for the gradual scoring system and are equivalent in quality to the old binary 70 threshold. The exit trigger fires when the score drops below 50 while

in position. No stop-loss, trailing stop, or time-based exit is implemented; the system relies entirely on score degradation for exit signals.

1.8 Trough Detection & Recovery Window

Trough detection uses `detect_troughs(series, rsi_col='RSI21', lookback=11)`: RSI21 must be ≤ 40 and be the minimum within an 11-bar centered window. Applied independently on all 3 timeframes. After each trough, the next 60 bars (`RECOVERY_WINDOW = 60`, reverted from 100) are marked as `in_recovery = True`. Only bars within the recovery window receive non-zero scores.

Engine 2: SENTINEL (v2.1)

The SENTINEL engine is a trough-gated recovery scoring system that operates as a downstream confirmation engine. It can be triggered by its own multi-timeframe trough detection OR by an upstream SCOUT engine signal (score ≥ 75). SENTINEL uses a unique 4-timeframe architecture including a synthesized 3-Day timeframe that bridges the gap between Daily and Weekly.

2.1 Scoring Formula

$$\text{core_raw} = \text{Pool_A} + \text{Pool_B} + \text{Pool_C} \quad (\text{capped at } 100)$$

$$\text{total_score} = \min(100, \text{core_raw}) + \text{Bonus_A} + \text{Bonus_B}$$

Theoretical maximum: $\min(100, 55+35+37) + 15 + 18 = 133$. The core pool cap at 100 prevents any single pool from dominating the total score.

Component	Max Points	Signals	Key Feature
Pool A (RSI Recovery)	55	8 x 3 TFs	Flat binary reversal + live/die crossovers
Pool B (Price Structure)	35	4 x 3 TFs	Binary: flatten, price > TP28, TP28 > WC50
Pool C (Confirmation)	37	7 signals	Divergence fire-and-hold + RSI > 50 gates
Bonus A (Structural)	15	3 signals	Volume surge + support bounce + 7-component trend quality
Bonus B (Uptrend)	18	5 signals	Gated by BB_Uptrend (MA200 rising + Close > MA200)
Core Cap	100	19 signals	$\min(100, A+B+C)$
Grand Maximum	133	24 signals	$100 + 15 + 18$

Table 5: SENTINEL scoring structure

2.2 Unique Feature: 4-Timeframe Architecture

SENTINEL is the only engine that uses 4 timeframes. The 3-Day timeframe is synthesized from Daily data by grouping every 3 daily bars into one (Open=first, High=max, Low=min, Close=last, Volume=sum). RSI, TP28, WC50, and MA200 are then recomputed from scratch on the synthesized series. This fills the gap between Daily (too noisy) and Weekly (too slow), providing smoother signals without Weekly's lag.

2.3 Unique Feature: Scout Gating

SENTINEL can be triggered by the SCOUT engine. When a SCOUT entry event has a score ≥ 75 , it creates a synthetic cycle in SENTINEL that opens the scoring window. Scout-triggered cycles use raised classification thresholds: Standard Entry requires ≥ 80 (vs 70 for trough-triggered), and Pilot Buy requires ≥ 55 (vs 50). This ensures that Scout-triggered entries meet a higher bar.

2.4 Unique Feature: Multi-TF OR-Gate Cycle Activation

Scoring is enabled on any daily bar where ANY of these conditions is true: Daily TF is in_recovery, 4H TF is in_recovery, 3D TF is in_recovery, Weekly TF is in_recovery, or a Scout-triggered cycle is active. This means a trough on any single timeframe opens scoring across ALL timeframes, maximizing the chance of catching early recovery signals.

2.5 Pool A: RSI Recovery (55 pts, 8 signals x 3 TFs)

Signal	4H	Daily	3D	Total	Type
A1: RSI Divergence	1.0	2.0	2.5	5.5	Fire-and-hold (multi-speed)
A2: RSI21 Reversal	2.0	4.0	5.0	11.0	Flat binary: RSI21 > cycle_low
A3: RSI36 Reversal	1.0	1.5	2.0	4.5	Flat binary: RSI36 > cycle_low
A4: RSI56 Reversal	0.5	1.0	1.5	3.0	Flat binary: RSI56 > cycle_low
A5: RSI21 > RSI36	2.5	4.0	5.0	11.5	Live/die crossover
A6: RSI21 > RSI56	1.5	2.5	3.5	7.5	Live/die crossover
A7: RSI36 > RSI56	1.0	2.0	2.5	5.5	Live/die crossover
A8: All RSI > 50	1.5	3.0	2.0	6.5	Live/die: all three above 50

Table 6: SENTINEL Pool A signals with per-TF weights

Note: SENTINEL's reversal signals (A2-A4) are flat binary, unlike SCOUT's gradual power-curve. This is a deliberate design choice: SENTINEL acts as a confirmation engine and uses simpler binary checks, while SCOUT provides the nuanced gradual scoring for early detection.

2.6 Pool B: Price Structure (35 pts, 4 signals x 3 TFs)

Signal	4H	Daily	3D	Total	Logic
B1: TP28 Flatten	2	1	1	4	TP28 range < 10% over last 10 bars
B2: WC50 Flatten	2	1	1	4	WC50 range < 10% over last 10 bars
B3: Close > TP28	3	4	6	13	Current close above TP28
B4: TP28 > WC50	3	5	6	14	TP28 above WC50

Table 7: SENTINEL Pool B signals with per-TF weights

2.7 Pool C: Confirmation (37 pts, 7 signals)

Signal	Max	Type	Logic
C_Div_D: Daily Divergence	8	Fire-and-hold	Daily multi-speed divergence active
C_Div_4H: 4H Divergence	5	Fire-and-hold	4H multi-speed divergence active
C_DRISI60: Daily RSI21 > 60	6	Live/die	Daily RSI21 exceeds 60
C_CrossTF: Cross-TF Alignment	6	Gated live/die	>=2 TFs with >=2 crossovers each + Price > TP28
C_DRISI50: Daily RSI21 > 50	5	Live/die	Daily RSI21 exceeds 50
C_WRSI50: Weekly RSI21 > 50	4	Live/die	Weekly RSI21 exceeds 50
C_4HRSI50: 4H RSI21 > 50	3	Live/die	4H RSI21 exceeds 50

Table 8: SENTINEL Pool C confirmation signals

2.8 Bonus Pools & Classification

Bonus A (15 pts) includes Volume Surge (5), Support Bounce with two-level peak analysis (3), and a 7-component Trend Quality score (10 points max: price position, higher lows, MA200 rising, performance vs MA200, and recovery rate). Bonus B (18 pts) is gated by BB_Uptrend (MA200 rising + Close > MA200) and includes Correction Depth (5), RSI Turning Positive (3), Price Above Support (3), and Volume on Recovery (4).

SENTINEL uses score-based entry/exit: ENTRY triggers at score crossing ≥ 75 from below; PILOT triggers at score crossing ≥ 50 . EXIT triggers when score drops below 40. Recovery window is 50 bars. Classification tiers: No Entry (<50), Pilot Buy (50-69), Standard Entry (70-79), Full Entry (80-89), Full Strong (90+).

Engine 3: APOLLO Analysis Engine

The APOLLO Analysis Engine is a trough-gated state machine designed specifically for Apollo Microsystems (NSE: APOLLOMICR). Unlike SCOUT and SENTINEL which use per-timeframe weighting, APOLLO uses implicit timeframe weighting through signal allocation. It is the only engine with stock-specific signals (multi-touch support zone hardcoded to 220-235 range) and a dedicated recovery analysis diagnostic tool.

3.1 Scoring Formula

$$\text{base_score} = \min(100, \text{Pool_A} + \text{Pool_B} + \text{Pool_C})$$

$$\text{total_score} = \text{base_score} + \text{Bonus_A} + \text{Bonus_B} \quad (\text{uncapped})$$

Component	Max Points	Signals	Key Feature
Pool A (Trough & Momentum)	42	6	Fire-and-hold troughs + recovery momentum windows
Pool B (Price Recovery)	25	5	Price > TP28/WC50, approaching TP28, higher lows
Pool C (Cross-TF Confirmation)	18	4	Trough agreement, RSI > 45/50 on multiple TFs
Bonus A (Volume & Holding)	8	3	Selling exhaustion + 4H volume expansion
Bonus B (Macro Context)	10	3	52-week room, multi-touch support, weekly reversal setup
Core Cap	100	18 signals	$\min(100, A+B+C)$
Grand Maximum	118	22 signals	$100 + 8 + 10$ (uncapped total)

Table 9: APOLLO Analysis scoring structure

3.2 Unique Feature: Cross-Timeframe Trough Matching

APOLLO's trough detection is more sophisticated than the other engines. When a daily trough is detected (RSI21 $\leq$ 40, local minimum in 11-bar window), the engine searches for a matching 4H trough within a plus or minus 3 calendar day window. The 4H trough uses a tighter threshold (RSI21 $\leq$ 38, local minimum in 5-bar window). Only when both daily and

4H troughs agree does the full scoring cycle activate, providing a stronger confirmation of the trough event.

3.3 Unique Feature: Recovery Momentum Windows

APOLLO measures recovery SPEED through time-limited windows: A3 (Daily Recovery Momentum) only fires within bars 1-10 after the trough AND requires RSI21 to have risen by at least 8 points from the trough level. A4 (4H Recovery Momentum) fires within bars 1-12 and requires a 10-point RSI21 rise. These windows create urgency: the engine rewards fast, strong recoveries rather than slow grinds.

3.4 Pool A: Trough & Recovery Momentum (42 pts, 6 signals)

Signal	Pts	Type	Logic
A1: Daily Trough Detected	10	Fire-and-hold	Always fires once daily trough exists (bars_since >= 0)
A2: 4H Trough Detected	8	Fire-and-hold	4H trough found within +/-3 days of daily trough
A3: Daily Recovery Momentum	8	Live/die	bars_since in (0,10] AND RSI21 rose >= 8 from trough
A4: 4H Recovery Momentum	6	Live/die	4H bars since in (0,12] AND 4H RSI21 rose >= 10
A5: 4H-RSI21 > 50	6	Live/die	Last 4H bar of the day has RSI21 > 50
A6: Multi-Speed Alignment	4	Live/die	Daily RSI21 > Daily RSI36

Table 10: APOLLO Pool A signals

3.5 Pool B: Price Recovery & Structure (25 pts, 5 signals)

Signal	Pts	Type	Logic
B1: 4H Close > 4H TP28	7	Live/die	Last 4H Close above last 4H Typical Price 28
B2: 4H Close > 4H WC50	6	Live/die	Last 4H Close above last 4H Wilder Close 50

Signal	Pts	Type	Logic
B3: Approaching D-TP28	5	Live/die	Daily Close within 3% below Daily TP28
B4: Higher Low Forming	4	Live/die	Current Daily Low > Trough Daily Low
B5: Price Above Trough +10%	3	Live/die	Close > Trough Low x 1.10

Table 11: APOLLO Pool B signals

3.6 Pool C & Bonus Pools

Signal	Pts	Pool	Logic
C1: Cross-TF Trough Agreement	6	C	4H trough matched (duplicates A2 confirmation)
C2: 4H-RSI36 > 45	4	C	Last 4H bar RSI36 exceeds 45
C3: Weekly RSI56 > 50	4	C	Last completed weekly bar RSI56 > 50
C4: Daily RSI21 > 45	4	C	Daily RSI21 exceeds 45
BA1: Selling Exhaustion	3	Bonus A	Trough day volume < 30-day avg (fire-and-hold)
BA2: Price Holding Above +10%	3	Bonus A	Close > Trough Low x 1.10 (same as B5)
BA3: 4H Volume Expansion	2	Bonus A	Any 4H bar in last 3 days > 1.5x 20-bar avg
BB1: 52-Week Room	3	Bonus B	365-day High > Current Close x 1.5
BB2: Multi-Touch Support (220-235)	4	Bonus B	>=2 separate touch clusters below 235 in 180 days
BB3: Weekly RSI Reversal Setup	3	Bonus B	W-RSI21 rising but still below W-RSI56

Table 12: APOLLO Pool C and Bonus pool signals

BB2 (Multi-Touch Support Zone) is APOLLO's only stock-specific signal, hardcoded to the 220-235 range based on historical support levels for APOLLOMICR. It counts the

number of separate touch clusters (consecutive days count as one cluster) where the daily low dropped to 235 or below within the last 180 trading days. BB3 (Weekly RSI Reversal Setup) detects the early stage of a weekly RSI crossover: W-RSI21 is rising (higher than previous week) but has not yet crossed above W-RSI56, indicating an imminent bullish crossover.

3.7 Entry, Exit & Classification

Entry threshold: score crosses ≥ 75 from below (not 70 as the 'Standard Entry' label might suggest). Pilot: score crosses ≥ 50 . Exit: score drops below 40 while in trade. Classification tiers match the other engines: No Entry (< 50), Pilot Buy (50-64), Standard Entry (65-79), Full Entry (80-89), Full Strong (90+). The effective entry bar is the first bar where `total_score` ≥ 75 , and the exit bar is the first subsequent bar where `total_score` < 40 .

Cross-Engine Comparison

The three engines form a layered detection system. SCOUT SPEC provides the most nuanced early detection with gradual scoring. SENTINEL acts as a downstream confirmation engine with multi-TF OR-gating and Scout triggering. APOLLO is a stock-specific state machine with cross-TF trough matching and recovery speed measurement. Together, they provide multiple independent perspectives on the same recovery opportunity.

Feature	SCOUT SPEC (v3.1)	SENTINEL (v2.1)	APOLLO Analysis
Total Signals	25	24	22
Core Cap	90	100	100
Grand Max	115	133	118 (uncapped)
Timeframes	4H, D, W	4H, D, 3D, W	4H, D, W
Recovery Window	60 bars	50 bars	No fixed window (state machine)
Entry Threshold	65 (Standard)	75 (ENTRY event)	75 (ENTRY event)
Exit Threshold	Score < 50	Score < 40	Score < 40
RSI Scoring	Gradual power-curve	Flat binary	Fire-and-hold + momentum windows
Price Scoring	Gradual (v3 redesign)	Binary	Binary
Trough Detection	Per-TF independent	Multi-TF OR-gate	Cross-TF matching (+/-3 days)
Scout Gating	No (is the upstream)	Yes (score >= 75)	No
Stock-Specific	No (multi-stock)	No (multi-stock)	Yes (hardcoded 220-235 support)
Unique Feature	WAP21 + gradual B-pool	3D synthesized TF	Recovery speed windows

Table 13: Cross-engine feature comparison

Signal Type Taxonomy

Across all three engines, signals fall into four behavioral categories that determine how they contribute to score stability. Understanding these categories is essential for diagnosing scoring volatility:

Gradual / Distance-Based: Score ramps linearly or via power-curve as a measured value changes. Examples: SCOUT A1-A3 (power-curve RSI recovery), SCOUT A5-A7 (spread-

based crossover), SCOUT B1-B5 (convergence, proximity, spread, WAP). These signals provide the smoothest scoring behavior and are the primary defense against volatility.

Fire-and-Hold: Triggered once at a specific event (trough detection, divergence confirmation) and persists for the entire cycle. Examples: SCOUT A4 (divergence), SENTINEL A1 (divergence), APOLLO A1/A2 (trough detection), APOLLO BA1 (selling exhaustion). These provide stable baseline points but can cause step-changes when they first fire.

Live/Die Binary: Re-evaluated every bar. Either fully on or fully off based on current conditions. Examples: SENTINEL A2-A8 (reversal, crossover), APOLLO B1-B5 (price structure), SCOUT C2 ($RSI > 50$). These are the primary source of scoring volatility when they flip on/off near threshold boundaries.

Binary Stepped: Multiple discrete levels rather than pure on/off. Examples: SCOUT A9-A10 (TF alignment counts), SCOUT BA1 (volume surge tiers), SENTINEL BA1 (volume tiers). These provide intermediate stability between gradual and binary.